National University of Sciences and Technology (NUST)

NUST Balochistan Campus (NBC)

CS-320 Artificial Neural Networks and Deep Learning

BS-AI 2024F · Fall 2026

Semester Project Proposal

Rebuilding AlexNet in PyTorch

Which of AlexNet's ingredients really matter, and does training from scratch still compete with ImageNet pretraining?

Project P01 · **AlexNet (NeurIPS 2012)**

Team Member	CMS ID	Role
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Project Proposal: Rebuilding AlexNet in PyTorch

CS-320 ANN and Deep Learning · Fall 2026 · Shahzaib Ali, Mubashara Anees, Abdul Ahad Khan
GitHub repository: https://github.com/shahzaibali-edu/Deep-Learning-Project

1. Paper and Datasets

Our paper is P01, *ImageNet Classification with Deep Convolutional Neural Networks* by Krizhevsky, Sutskever and Hinton (NeurIPS 2012). It trains an eight-layer CNN (five convolutional and three fully connected layers, 60M parameters) on 1.2M ImageNet images; a single network reached 37.5% top-1 and 17.0% top-5 error on ILSVRC-2010, and an ensemble won ILSVRC-2012 with 15.3% top-5 error against 26.2% for the runner-up. The gain came from a bundle of practical ideas: ReLU instead of tanh, GPU training, local response normalisation (LRN), overlapping max-pooling, dropout and data augmentation.

We will use CIFAR-10 for the paper's ReLU-vs-tanh experiment (Figure 1), Imagenette (10 ImageNet classes) in place of full ImageNet, and Oxford-IIIT Pets (37 breeds, CC BY-SA 4.0) as the new dataset the paper never used, all on free Colab (T4). We will show all three loading in Colab.

2. Problem Statement

AlexNet bundles many tricks and took five to six days on two GPUs to train. It is unclear which tricks matter at small scale, and whether training from scratch on a small dataset can compete with transfer learning. We will rebuild it, reproduce its key results on Colab, and ablate it.

3. Research Questions

RQ1. Does ReLU really train faster? How many epochs does the paper's four-layer CNN need to reach 25% training error on CIFAR-10 with ReLU versus tanh, and is our speed-up close to the paper's six-fold?

RQ2. Can AlexNet be reproduced at Colab scale? What top-1 and top-5 accuracy does an AlexNet-style network reach on Imagenette, and how much of the gap to the paper's ImageNet results comes from data size, training length and the single GPU?

RQ3. Which ingredients matter most? From one fixed baseline (same seed and split), how much does switching off dropout, overlapping pooling, LRN, augmentation or weight decay change validation accuracy and the train–validation gap?

RQ4. Scratch or pretrained? On Oxford-IIIT Pets, how much better is an ImageNet-pretrained AlexNet (torchvision) than the same network trained from scratch, and where do their errors and first-layer kernels (the paper's Figure 3) differ?

4. Approach

- **Read and explain:** architecture diagram with tensor shapes at every layer and a hand parameter count (e.g. conv1: $96 \cdot 11 \cdot 11 \cdot 3 + 96 = 34{,}944$).
- **Reproduce:** Figure 1 on CIFAR-10, then an AlexNet-style network on Imagenette (SGD, momentum 0.9, weight decay 5×10^{-4}, batch 128, mixed precision).
- **Extend and ablate:** Pets with a small-CNN baseline, scratch vs. pretrained, five ablations, error analysis.
- **Debug and share:** the PyTorch checklist (first loss near $\ln C$, overfit one batch, gradient checks, Drive checkpoints), a bug diary, a Gradio demo and the one-pager.

5. Role Split and Plan

Shahzaib (paper lead) owns architecture and reproduction and implements the convolutional stack with LRN and pooling. Mubashara (data lead) owns datasets, shapes and evaluation and implements the data pipeline with augmentation. Abdul Ahad (experiments lead) owns ablations, demo and report and implements the classifier head with dropout and the training loop. Each member runs at least one ablation and commits to our shared GitHub repository. Any generative-AI help will be disclosed.